Fiscal Year:	FY 2011	**Task Last Updated:**	FY 10/30/2011
PI Name:	Dingfelder, Michael Ph.D.		
Project Title:	Patterns of Energy Deposition by HZE Particles in Cellular Targets		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	27858	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
Start Date:	05/15/2004	**End Date:**	07/31/2011
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
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Flight Program:			
Flight Assignment:	NOTE: Change in Risk/Gaps per HRR information (Ed., 9/23/2011) NOTE: New end date is 7/31/2011 per C. Guidry/JSC (6/2010) NOTE: New end date is 7/31/2010, per PI (9/2009) NOTE: Received NCE to 7/31/2009 per PI (1/2009) NOTE: End date changed to 12/31/2008 per S. Krenek/JSC (8/07) End date changed to 7/31/2008 per JSC info update (10/06)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Toburen, Larry (East Carolina University)		
Grant/Contract No.:	NNJ04HF39G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The interpretation of data from mechanistic studies on radiation biology requires detailed descriptions of the local patterns of energy deposition in the appropriate cellular or sub-cellular sites to explore the effects of radiation quality on biological response. The objective of this project is to provide Monte Carlo (MC) simulations on the local patterns of energy deposition by HZE particles in the appropriate cellular environment including effects of micro-, and macro-geometrical considerations. The calculations focus on interpretation of microscopic “event” or “dose” distributions appropriate to recently initiated studies of HZE radiation induced leukemia in animal and cellular models [NSCOR Program, Colorado State University, R. Ulrich, PI] and those encountered in the production of clustered damage in DNA [e.g., Brookhaven National Laboratory, B Sutherland, PI]. The progenitor cells for leukemia are haematopoietic stem cells that are located in clusters within marrow cavities having dimensions on the order of a few hundred microns. These cavities are surrounded by trabecular bone matrix, which forms a wall of several hundred microns thickness around the marrow. The dimensions and geometry of the bone marrow are too irregular for direct dosimetry measurements and dose-averaging methods are inappropriate for chronic low dose-rate (single particle) exposures. Detailed Monte Carlo charged particle track structure simulation will provide quantitative information on spatial patterns of energy deposition in cellular targets of dimensions as small as a few nanometers on a particle-by-particle basis; these distributions are the starting point to characterize the detailed nature of radiation-induced damage produced by high-LET radiation. The modeling approach will combine the capabilities of full event-by-event Monte Carlo track simulation codes with codes employing condensed history capabilities; the latter is more appropriate for the description of high-energy secondary electrons. The event-by-event ion code PROTRAC (which is part of the biophysical simulation tool PARTRAC) developed at GSF Neuherberg for protons and alpha particles will be extended for heavy charged particle (HZE) interactions using appropriate relativistic theory. Generated secondary electrons that traverse specific regions of interest, e.g., bone marrow cells or their nuclei, will be followed also event-by-event using ETRAC (the electron transport code of PARTRAC) whereas electron transport in adjoining tissues and the surrounding structural material will be followed by the more computer efficient condensed history general purpose MC transport code PENELOPE developed at the University of Barcelona. Appropriate interaction cross sections for a wide range of target materials will be added to the ETRAC and PROTRAC codes, specifically to explore the effects of bone-tissue interfaces on energy deposition in bone marrow cells. Comparison of the spatial pattern of energy deposited in volumes of interest obtained from models of the HZE particles with that obtained from low-LET radiation will provide a firm basis to interpret radiation quality effects on biological response and the mathematical structure developed is amenable to scaling to volumes appropriate to human dimensions for considerations of risk assessment. The objective of this project is to provide Monte Carlo (MC) simulations on the local patterns of energy deposition by HZE particles in the appropriate cellular environment including effects of micro-, and macro-geometrical considerations. The focus will be on Leukemia, i.e., on the bone/bone-marrow interface from realistic mouse or human dimensions down to single cells. Little is known on the interaction of HZE particles with the human body. HZE particles are part of the Galactic Cosmic Rays (GCR) which exist in deep space and cannot be shielded completely inside spacecrafts. The latency time of leukemia, 2-3 years, coincides more or less with the mission time for the planned expedition to Mars.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	
Task Progress:	During this final no-cost extension period we are finishing up working on the interaction cross sections for metallic calcium with charged particles. One of the main issues was the modeling of the energy loss function. While using standard techniques and a Gaussian model we had problems in calculating cross sections at highly relativistic energies. It turned out this problems were caused by the incorrect asymptotic behavior of the Gaussian functions. We therefore modified our model and the model functions. We have now finished the modeling of the energy loss function and have calculated total and energy differential IMFPs for electron, proton, and alpha particle impact. Stopping powers, and total cross sections now coincide with the few known experimental/theoretical data. We are currently implementing the cross sections into our Monte Carlo codes and plan to rerun some of the reported dosimetric simulations on energy depositions in trabecular bone. However, we do not expect to see any differences in outcome to the simulations already performed using density scaled water instead of metallic calcium as bone surrogate, at least not at high relativistic energies.
Bibliography Type:	Description: (Last Updated: 11/09/2018)
Abstracts for Journals and Proceedings	Dingfelder M. "Track structure simulations at relativistic energies: an update on cross section calculations." Invited talk given at the 38th COSPAR Scientific Assembly, Bremen, Germany, July 18-25, 2010. 38th COSPAR Scientific Assembly, Bremen, Germany, July 18-25, 2010. Book of Abstracts, F22-0018-10. , Jul-2010
Abstracts for Journals and Proceedings	Dingfelder M. "Current developments and challenges of ion track structure simulations." Invited talk given at the - An international workshop in Monte Carlo computational methods in radiation track simulations and applications in physical, biological, and medical sciences, Stockholm, Sweden, November 9-12, 2010. MC2010 Stockholm, November 2010. Book of Abstracts. , Nov-2010
Abstracts for Journals and Proceedings	Dingfelder M. "Track structure simulations for charged particles." Invited talk given at the 2011 NCRP (National Council on Radiation Protection & Measurements) 47th Annual Meeting on 'Scientific and Policy Challenges of Particle Radiations in Medical Therapy and Space Missions', Bethesda, MD, March 7-8, 2011. 47th Annual Meeting on 'Scientific and Policy Challenges of Particle Radiations in Medical Therapy and Space Missions', Bethesda, MD, March 7-8, 2011. Book of Abstracts. p. 16-17. http://www.ncrponline.org/Annual_Mtgs/2011_Ann_Mtg/2011_NCRP_Annual%20Meeting.pdf , Mar-2011
Abstracts for Journals and Proceedings	Jorjishvili IG, Dingfelder M, Toburen LH. "The generalized oscillator strength of metallic calcium, and the mean free path of and the stopping power for charged particles in calcium." Poster presented by I.G. Jorjishvili at the 56th Annual Meeting of the Radiation Research Society, Maui, HI, September 25-29, 2010. 56th Annual Meeting, Radiation Research Society, September 25-29, 2010. http://www.abstractsonline.com/Plan/ViewAbstract.aspx?mID=2569&sKey=df5d6e42-792b-4262-8956-74eba4f8daa0&cKey=7e5c6c18-f8e6-4cdc-ae08-181c7ea804b2&mKey=5fb93787-2553-4b3b-a915-8c1ae71f127a , Sep-2010
Abstracts for Journals and Proceedings	Dingfelder M, Jorjishvili IG. "Heavy ion track structure simulations in liquid water and bone." Poster presented at the 14th International Congress on Radiation Research, Warsaw, Poland, August 26 - September 1, 2011. 14th International Congress of Radiation Research, August, 2011. Book of Abstracts, POS06-01, p. 88. , Aug-2011
Articles in Peer-reviewed Journals	Friedland W, Dingfelder M, Kundrát P, Jacob P. "Track structures, DNA targets and radiation effects in the biophysical Monte Carlo simulation code PARTRAC." Mutation Research. 2011 Jun 3;711(1-2):28-40. Epub 2011 Jan 31. Review. PubMed PMID: 21281649 , Jun-2011
Articles in Peer-reviewed Journals	Fernández-Varea JM, Segui S, Dingfelder M. "Lalpha, Lß, and Lgamma x-ray production cross sections of Hf, Ta, W, Re, Os, Au, Pb, and Bi by electron impact: Comparison of distored-wave calculations with experiment." Phys Rev A. 2011 Feb;83(2):022702. http://dx.doi.org/10.1103/PhysRevA.83.022702 , Feb-2011